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Basics of X-ray Diffraction

H. STANJEK[1] and W. HÄUSLER[2]
[1] *Institut für Mineralogie und Lagerstättenlehre, Rheinisch Westfälische Technische Hochschule, 52056 Aachen, Germany*
[2] *Physik-Department E15, Technische Universität München, 85747 Garching, Germany*

Abstract. X-ray diffraction (XRD) is the most comprehensive tool to identify minerals in complex mineral assemblages. The method is briefly described with special emphasis on clay and ceramics. As an example, an investigation of graphite-containing pottery sherds by XRD is presented. By comparing the measured XRD data with the patterns simulated by the Rietveld method, the graphite content of such samples could be determined.

Key words: X-ray diffraction, Rietveld simulation, graphite, graphite clays, black pottery.

1. Introduction

X-ray diffraction (XRD) is an important tool in mineralogy for identifying, quantifying and characterising minerals in complex mineral assemblages. Its application to ancient ceramics, which are a mixture of clay minerals, additive minerals and their transformation products yields information on the mineral composition of objects. Details of production processes, like firing temperatures and kiln atmospheres as well as applications of slips or glazes may thus become transparent.

This chapter gives some basic background information about the physics of the X-ray diffraction process and its application to pottery clays and ceramics. While it is relatively easy to determine which minerals a specimen contains from the positions and rough intensities of the diffraction peaks, it is much more difficult to give the contents of individual minerals quantitatively, because for the latter one needs to model the intensities of the peaks in the X-ray diffraction pattern accurately. This is a difficult task into which many parameters enter. It is, however, within the reach of computer based approaches such as Rietveld analysis [1], the basic concepts of which will be described. As an example we have chosen an X-ray study of pottery sherds in which graphite was tentatively identified by scanning electron microscopy [2]. The task of XRD was therefore to identify this mineral and, if possible, to quantify the content of it in individual sherds.

Before we outline the theory of X-ray diffraction, two concepts need to be explained: what is a unit cell and how can diffracting lattice planes be handled in a convenient way. Crystalline minerals – only few minerals like obsidian are not crystalline – are uniquely characterised by their chemical composition and

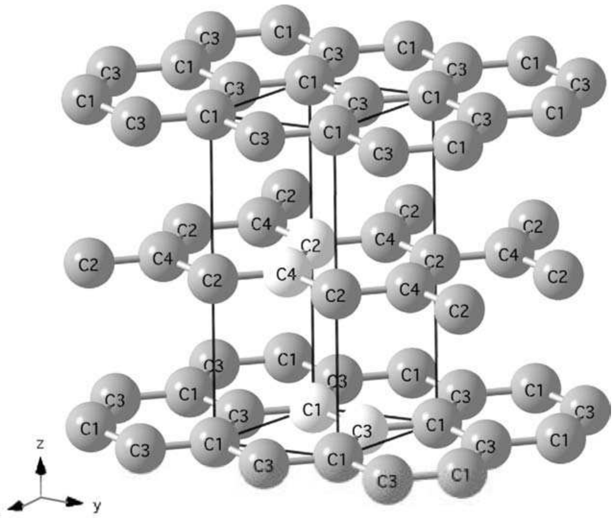

Figure 1. The hexagonal structure of graphite: Carbon sheets built from rings of 6 carbons atoms stack along the z axis. The unit cell (solid lines) contains 4 atoms (plotted in lighter shades). The carbon atom C1, for instance, has the coordinates $(0, 0, 0)$, C2 $(0, 0, 1/2)$, C3 $(1/3, 2/3, 0)$ and C4 $(2/3, 1/3, 1/2)$.

the three-dimensional arrangement of atoms in their structure [3]. The smallest unit that contains all the necessary structural and chemical information to uniquely define a mineral is called a unit cell (Figure 1). A macroscopic crystal then consists of a three dimensional, ordered arrangement of such unit cells.

The sets of three coordinates mentioned in Figure 1 refer to the coordinates of the individual atoms in the unit cell along the x, y, and z axis. The unit cell has the dimensions a, b and c along these axes. The position of the carbon atom C3, for instance, at $(1/3, 2/3, 0)$ is obtained by starting from the origin $(0, 0, 0)$ and going $1/3$ of the unit cell dimension a along the x axis and $2/3$ of b along the y axis, while z stays at zero. These coordinates are called fractional. For determining absolute distances, the geometry of the unit cell, i.e., the lengths of the cell dimensions a, b and c and the angles between the axes must be known. For graphite, $a = b = 2.456$ Å, $c = 6.696$ Å, and the x axis intersects the y axis at an angle of 120° while the z axis is perpendicular to both x and y. The absolute distance between the two carbons at $(0, 0, 0)$ and $(1/3, 2/3, 0)$ is then 1.418 Å.

The representation of planes of atoms is also straightforward. First, the intersections of the lattice plane with the x, y, and z axis are determined. For reasons that need not be explained here, the reciprocal values of the coordinates of the intersections are calculated and then converted to integer numbers h, k and l by appropriate multiplication. The hkl triples are called the Miller indices of a plane or a face. An example: a plane intersects the x axis at $1/3\,a$, the y axis at $1/2\,b$ and the z axis at $2\,c$. The reciprocals are then (3, 2, 1/2), which, by multiplication with 2 to obtain integer values, finally gives (641). Planes parallel to an axis intersect at infinity, the reciprocal of which is zero. A (100) plane is therefore parallel to the y and z axes and intersects the x axis at $1\,a$.

2. Theory of diffraction

The interaction of waves with periodic structures produces diffraction effects if the wavelength and the periodicity of the crystals, are of similar magnitude. X-rays may easily be produced with wavelengths matching the unit cell dimensions of crystals, but electrons or neutrons of appropriate energy can also be used for diffraction experiments on crystals.

Considering that atoms have diameters of the order of Ångströms (1 Å $= 10^{-10}$ m), unit cells have dimensions of several Å. This implies that crystals with sizes of microns or larger consist of billions of unit cells, which repeat periodically in all three dimensions, i.e., they possess long-range order. This kind of order distinguishes crystalline materials from amorphous ones, e.g., glasses, which have only short-range order. Since the "quality" of diffraction effects in XRD depends strongly on the strict and undisturbed periodicity of atoms, any kind of deviation from the ideal order will show in the X-ray diffraction diagram. Even small crystallite size is a deviation from the theoretically infinite perfect crystal. Other deviations from the ideal order may be replacements of atoms by others (common in solid solutions), slight geometric deviations of atoms from their ideal position due to internal strain (e.g., from incomplete annealing or massive grinding), or larger two or three dimensional aberrations (e.g., dislocations or stacking faults).

2.1. THE ATOM FORM FACTOR

Electromagnetic waves with wavelengths of the order of 10^{-10} m are called X-rays. The electric field of such waves interacts with the charges of all electrons of an atom, which then emit an almost spherical wave with the same wavelength as the incident radiation. The amplitude of this outgoing wave is proportional to the number of electrons in the atom, and, hence, to the atomic number. Light elements with few electrons, e.g., carbon or oxygen, are therefore "poor" scatterers for X-rays, whereas heavy elements such as lead are "good" scatterers. Detection limits are severely influenced by this effect. The amplitude of the scattered wave is described by the atom form factor f. Due to interference within individual atoms, especially larger ones, the amplitude of the outgoing wave and hence the atom form factor varies also with the scattering angle 2Θ (Figure 2).

2.2. DIFFRACTION AND BRAGG'S EQUATION

Without any diffraction effects, the incidence of a primary X-ray beam onto a sample volume would produce scattering in all directions. Diffraction redistributes intensity from the whole scattering sphere into distinct directions. Therefore,

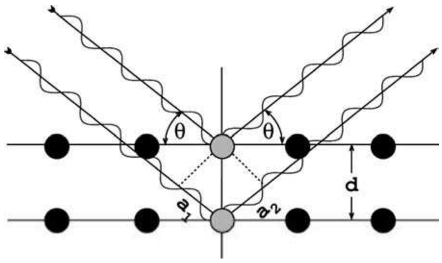

Figure 2. Geometric derivation of Bragg's law: Constructive interference occures when the delay between waves scattered from adjacent lattice planes given by $a_1 + a_2$ is an integer multiple of the wavelength λ.

intensity peaks arise in certain directions, whereas in directions between peaks the intensity decreases drastically. The intensity integrated over the sphere, however, remains constant due to energy conservation. In what directions do we observe such peaks, also called reflections? One way of describing these directions is the notion of scattering lattice planes and interference between the wavelets scattered by neighbouring lattice planes. Figure 2 illustrates this situation. Constructive interference and hence a so called Bragg reflection is obtained when the path of the wavelet scattered of the lower of the two planes is longer by an integer number of wavelengths λ than that of the wavelet scattered off the upper plane. A reflection will thus occur when

$$n\lambda = 2d \sin\Theta. \tag{1}$$

This is the so-called Bragg equation, where λ is the wavelength of the radiation, n is an integer number, Θ is the angle between the lattice planes and the incident beam and d is the distance of the lattice planes for which the peak occurs.

2.3. THE INTERFERENCE FUNCTION

One might argue that small deviations from the ideal Bragg angle Θ (Figure 2) should also produce significant intensity, as long as the phase delay is not far from λ and therefore constructive interference still occurs to some extent, although not at maximum intensity. Indeed, this is the case in sufficiently small crystals. In large crystals, whenever at an angle 2Θ the phase delay is not exactly λ, a position somewhere in the crystal can be found that gives rise to a phase delay of $\lambda/2$ and hence destructive interference. Therefore, in large crystals even minor deviations from the ideal Bragg angle lead to cancellation by interference and therefore sharp peaks result. For small crystals, however, the peaks broaden. This influence of crystal size is modelled by the interference function

$$S = \frac{(\sin \pi h N)^2}{N(\sin \pi h)^2}, \tag{2}$$

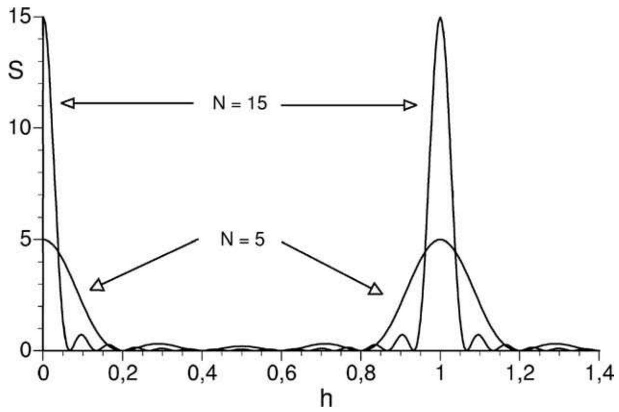

Figure 3. Plot of interference function S against the continous variable h for $N = 5$ and $N = 15$. Note that the area under the peaks remains constant irrespective of N.

where N is the number of unit cells in the crystal and h the index of the reflection (see below). In natural samples, the ripples occurring at non-integer values of h (Figure 3) are never observed, because minerals always have a particle size distribution, which smooths the ripples. The interference function is calculated individually for the directions along x, y, and z, respectively, and the results multiplied to give the interference function Y for the crystallite.

2.4. THE STRUCTURE FACTOR

As a next step we have to put the diffraction by crystals on a quantitative basis in order to understand why different kinds of structures, i.e., different mineral phases show unique sets of diffraction peaks and why these peaks have characteristic relative intensities. Two quantities need to be considered, when waves interact: their amplitude and their relative phase. Both parameters make up a complex number. The process of interference of waves scattered by atoms at different positions within a unit cell and with different amplitudes reduces to simply adding all these complex numbers to give the structure factor F as

$$F(hkl) = \sum_n f_n \exp(-\mathrm{i}\Phi_n) = \sum_n f_n(\cos\Phi_n + \mathrm{i}\sin\Phi_n). \quad (3)$$

The summation goes over all n atoms in the unit cell and the f_n are the atom form factors of these n atoms. Φ_n is given by

$$\Phi_n = 2\pi(hx_n + ky_n + lz_n) \quad (4)$$

in which x_n, y_n, and z_n are the fractional coordinates of atom n within the unit cell and h, k and l are the Miller indices of the respective set of lattice planes giving rise to the reflection. Equation (3) gives the amplitude of a diffracted wave, but experimentally we observe the intensity, which is proportional to $|F(hkl)|^2$.

As an example, we calculate the structure factor of the 002 reflection of graphite. Since all atoms are identical, we may set f_n equal to unity as long as we are not interested in absolute intensities. Furthermore, since $h = k = 0$ and $l = 2$, Equation (4) reduces to $\Phi = 2\pi \cdot 2z = 4\pi z$. Inserting this into Equation (3) yields

$$
\begin{aligned}
F(002) &= \cos(4\pi \cdot 0) + \mathrm{i}\sin(4\pi \cdot 0) + \cos\left(4\pi \cdot \frac{1}{2}\right) + \mathrm{i}\sin\left(4\pi \cdot \frac{1}{2}\right) \\
&\quad + \cos(4\pi \cdot 0) + \mathrm{i}\sin(4\pi \cdot 0) + \cos\left(4\pi \cdot \frac{1}{2}\right) + \mathrm{i}\sin\left(4\pi \cdot \frac{1}{2}\right), \\
&= 1 + 0 + 1 + 0 + 1 + 0 + 1 + 0, \\
&= 4.
\end{aligned}
$$

Recalculating the example for 001 shows that $F(001) = 0$. This means that the 001 reflection does not occur because of extinction. This occurs because between the waves reflected by the hexagonal planes Figure 1 containing atoms C1 and C3 and those reflected by planes at $c/2$ containing atoms C2 and C4 there is a phase shift of $\lambda/2$. The two sets of waves therefore cancel by destructive interference.

2.5. THE LORENTZ AND THE TEMPERATURE FACTOR

Our aim to model the XRD pattern of a mineral requires us to consider additional factors. One of these is the Lorentz factor, which takes into account that in a powder the amount of crystals that contribute to the measurable intensity varies with the diffraction angle Θ. The Lorentz factor is frequently combined with the polarisation factor [4]. Moreover, the atom form factor f, introduced in Section 2.1 requires a structure-dependent modification. With increasing temperatures the atoms vibrate more and more strongly around their ideal positions. Hence, their power to scatter waves is slightly reduced and this reduces the intensities, again dependent on Θ. This effect is described by temperature factors for each kind of atom in the unit cell.

2.6. EXPERIMENTAL FACTORS

The X-rays penetrate the sample to some extent. It is therefore necessary to have an “infinetely” thick sample, where the contribution to the intensity from the rear side of the sample is negligible. In practice this means a sample loading of more than about 20 mg/cm^2. For thinner samples, the peak intensity is reduced from the intensity I_0 for an infinitely thick sample to intensity I according to

$$\frac{I}{I_0} = 1 - \exp\left(-\frac{2\mu^* g}{\sin\Theta}\right), \tag{5}$$

where μ^* is the effective mass absorption coefficient in cm^2/g [5] and g is the sample thickness in units of g/cm^2. The product $\mu^* g$ can be measured easily by

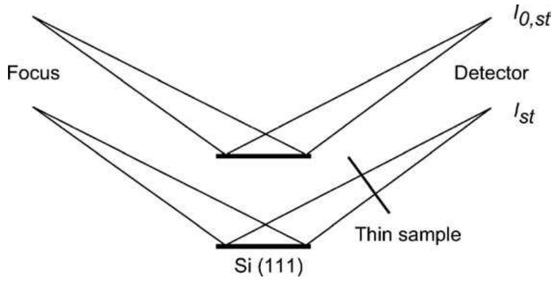

Figure 4. Determination of the mass absorption of a thin sample. First, 2Θ is set on a strong peak of a Si (or quartz) plate and $I_{0,\mathrm{st}}$ is measured. After inserting a thin sample into the beam I_{st} is measured. See Equation (6) for evaluating $\mu^* g$.

putting a silicon or quartz specimen as an intensity standard into the goniometer. The intensity $I_{0,\mathrm{st}}$ of a strong peak of this standard at a Bragg angle Θ_{st} is then measured. After placing the sample for which $\mu^* g$ is to be determined above the standard into the goniometer, the now reduced intensity I_{st} of the standard peak is measured (Figure 4). The ratio of both count rates yields $\mu^* g$ according to

$$\mu^* g = -\ln(I_{\mathrm{st}}/I_{0,\mathrm{st}}). \tag{6}$$

Entering $\mu^* g$ into Equation (5) then allows a point-by-point correction of the counts obtained with the thin sample to a theoretical, infinitely thick sample.

2.7. THE RIETVELD METHOD

When studying complex mixtures of minerals, one would like to use X-ray diffraction not only to determine which minerals a sample contains, but also to give the relative concentrations of the individual minerals. Rietveld programs [1] use all factors mentioned above to simulate the X-ray diffraction pattern of a sample composed of a mixture of minerals. For each mineral, the Bragg angles and the intensities must be calculated. Then the total experimental diffraction pattern can be modelled by adjusting the concentrations of the individual components appropriately. In this approach the user first has to identify all minerals in a sample, a task which need not be done by using the Rietveld routine. Then, all the necessary structural information (symmetry and dimensions of the unit cell, positions of atoms) has to be gathered from existing data bases and entered into a starting file. After guessing the approximate composition of the sample, rough estimates of the profile parameters that determine the width and shape of the individual Bragg reflections are needed. With all this information, the Rietveld program is able to simulate the XRD pattern from the given starting parameters. This simulation, however, will deviate more or less from the measured scan. The purpose of Rietveld programs is, therefore, to tune individual parameters such as the content of individual minerals, the profile function parameters or the unit cell dimensions to obtain better agreement between simulation and measurement. This optimisation is called fitting and is done in iterative cycles, because we deal with a highly non-linear system to be optimised.

Rietveld programs need to know, how "good" a fit is. This can be calculated by adding up the squared differences between the measured diffraction pattern and the simulated one. By normalising this figure, one obtains a measure called Bragg's R_{wp}, which tells us how "good" the current parameter set is able to simulate a measured pattern. The smaller the R_{wp}, the better. It is not possible, however, to find a parameter set for which $R_{wp} = 0$ (i.e., perfect agreement between simulation and measurement), because the counting statistics in measured scans limit R_{wp} to a minimum value called R_{exp}. The ratio of R_{wp}/R_{exp}, called the goodness-of-fit, gives a good indication about the obtained optimisation level relative to the theoretically best one.

3. An example: graphite in ancient ceramics

Graphite is sometimes found as a constituent of ceramics. It causes a black and sometimes even shiny appearance [6]. Rarely it was added as a mineral by the ancient potters. More often, non-crystalline carbon black seems to have deposited on ceramics in the kiln and may have transformed at least partly into crystalline graphite. The identification of crystalline graphite in ceramics by XRD is a fine example where not only the presence of peaks has to be considered, but also their relative intensities need to be taken into account quantitatively, not only if one wishes to determine the graphite concentration quantitatively, but even to be able to identify graphite in the ceramic material at all. The major problem with graphite in ceramics is that its best diagnostic peak, the 002 reflection, coincides with the most intense peak 101 of the omnipresent quartz (Table I). The unambiguous identification of graphite is further hampered by low intensities of the other graphite peaks except 101 with $I_{rel} = 17\%$. The 101 reflection of graphite at 2.03 Å is not overlapping with any quartz peak, but with peaks of other minerals common in ceramics, such as phlogopite and biotite (Table I). One therefore needs to take into account the intensities of the reflections for a unique identification of graphite in ceramics.

3.1. EXPERIMENTAL DETAILS

3.1.1. *Sample description*

Samples 19/936 and 19/652 stem from the Celtic oppidum of Manching, Germany [7]. Sample 39/323 is a black replica pottery produced during field firings in Huaca Sialupe, Peru [6]. Samples 19/936 and 19/652 were made from so called graphite clays, which are man-made mixtures of ordinary pottery clays with graphite, which in the present case came from the graphite deposit near Kropfmühl, Lower Bavaria, Germany. The sample of black replica pottery (39/323) is a piece of ceramics with surface carbon introduced during the firing process by soot deposition. Its black

Table I. d values, hkl indices and relative intensities of XRD-peaks of quartz and muscovite, which are close to graphite peaks

Quartz			Muscovite			Graphite		
d [Å]	hkl	I_{rel}	d [Å]	hkl	I_{rel}	d [Å]	hkl	I_{rel}
4.254	100	27						
3.343	**101**	**100**	**3.339**	**024**	**60**	**3.348**	**002**	**100**
			3.338	**006**	**65**			
2.456	110	11	2.137	135	49	2.127	100	3
2.281	102	2	2.123	043	4	2.027	101	17
2.236	111	4	2.044	044	12	1.795	102	3
2.127	200	8				1.674	004	6
1.979	201	5				1.539	103	5

and shiny appearance has been enhanced by polishing the ceramic surface prior to the firing process with special polishing stones. Soot may deposit on the polished clay in form of graphite. In the replica ceramics, hexagonal graphite platelets were observed by scanning electron microscopy (SEM) [2, 8]. Total carbon analyses give 115 ± 2 mg $C \cdot g^{-1}$ for the graphite ware 19/936 from Manching, 110 ± 1 mg $C \cdot g^{-1}$ for 19/652 and 41 ± 1 mg $C \cdot g^{-1}$ for the black replica pottery, 39/323. For XRD analyses, the samples were ground under acetone and sieved to <20 μm. Larger particle sizes must be avoided, since otherwise the number of particles in the specimen used for the diffraction experiments becomes too low to give reproducible intensities. The X-ray scans were run on a Philips PW 1040 goniometer using Co K_α radiation. The divergence slit of the goniometer was chosen small enough to avoid beam overflow, which would decrease the intensities of peaks at smaller diffraction angles 2Θ, when some of the primary beam tends to pass by the sample holder.

The precision with which mineral phases can be quantified and characterised increases with the improvement of the counting statistics. All scans were therefore step-scanned with 10 seconds counting time per step to give maximum intensities of the strongest quartz peak of more than 60 000 counts. The angle increment was chosen as 0.02 °2Θ to give more than 7 data points for the sharp profiles of quartz. Total scan times were therefore half a day.

Samples 19/652 and 19/936 were fitted with the RIETAN computer program [9]. The parameters that were refined are the zero angle of the instrument, the background and for each phase the concentration, the profile parameters of the modified Pseudo-Voigt function [10], the unit cell parameters (except for quartz) and the particle orientation function. All other structural parameters (positions of atoms, temperature factor, site occupancies) were held constant.

In the case of the black replica pottery 39/323 the amount of sample material was insufficient to give a thick specimen. A thin sample was prepared by depositing a slurry of the finely ground sample material dispersed in ethanol onto a mylar film. The absorption was measured as outlined in Section 2.6 and the scanned pattern recalculated using Equation (5). This scan was fitted with the RIETAN program [11] in the regions of 19–36 °2Θ and 42–44 °2Θ with the relative line intensities fixed for each phase.

3.2. RESULTS

In the graphite ware specimen 19/936, quartz, mica, K-feldspar and an intermediate plagioclase were identified as main components by the MacClayFit procedure. The intensity ratio of the two quartz peaks 100 and 101 deviated, significantly from its ideal ratio of 0.187 (cf. Table I), which indicates the presence of graphite, the 002 of which overlaps almost completely with the 101 of quartz. The fit from 14–82 °2Θ with additional graphite converged at $R_{wp} = 11.97\%$ with a goodness-of-fit (*GOF*) of 1.96 (Figure 5). The refined contents were: graphite 0.13(2), quartz 0.31(1), muscovite 0.35(1), microcline 0.13(1) and anorthite 0.09(1) g/g. An inspection of the residuals shows that peaks have not been fitted well at 15.4, 20.5 and 23 °2Θ (Figure 5). The peak at 15.4 °2Θ $\widehat{=}$ 6.71 Å would correspond to the 001 peak of graphite, which, has zero intensity in ideal graphite. Irregular stacking of the graphite layers (see Figure 1), however, would inhibit perfect extinction of the 001 reflection. Hence, this broad peak indicates a poor quality of the graphite.

The mineralogy of sample 19/652 is similar to that of sample 19/936. The fit converged at $R_{wp} = 12.1\%$ and a *GOF* of 1.93. Refined contents were: graphite 0.087(36), muscovite 0.50(1), anorthite 0.13(1), microcline 0.056(5) and hematite 0.048(4) g/g. For graphite, a pronounced anisotropic peak broadening yielded a mean coherence length whereas the coherence length perpendicular to *c* of only 6 nm, whereas the coherence length along the *c* axis was found to be 850 nm. This indicates considerable stacking faults.

The black replica pottery contained quartz and muscovite as the main phases plus additional phases such as feldspars, which were not identified in detail but fitted as a free phase. The fit without graphite converged to $R_{wp} = 6.206\%$ and a *GOF* of 1.77, whereas the addition of the 002 peak of graphite improved the fit quality slightly to 6.087% and a *GOF* of 1.74. Although the number of refined parameters were 88 and 91, respectively, this improvement is far from significant according to the F-test, which describes the standard deviations. Furthermore, the Durbin–Watson d statistics indicated severe serial correlations between consecutive residuals, which invalidates the comparison of the Bragg R's [12]. Considering that the halfwidth parameters of the graphite peak resulted to be unrealistically low – graphite would have a 002 peak even sharper than that of quartz – we conclude that graphite could not be identified unambiguously in this sample. Its content was below the detection limit either because of the relatively low total carbon content

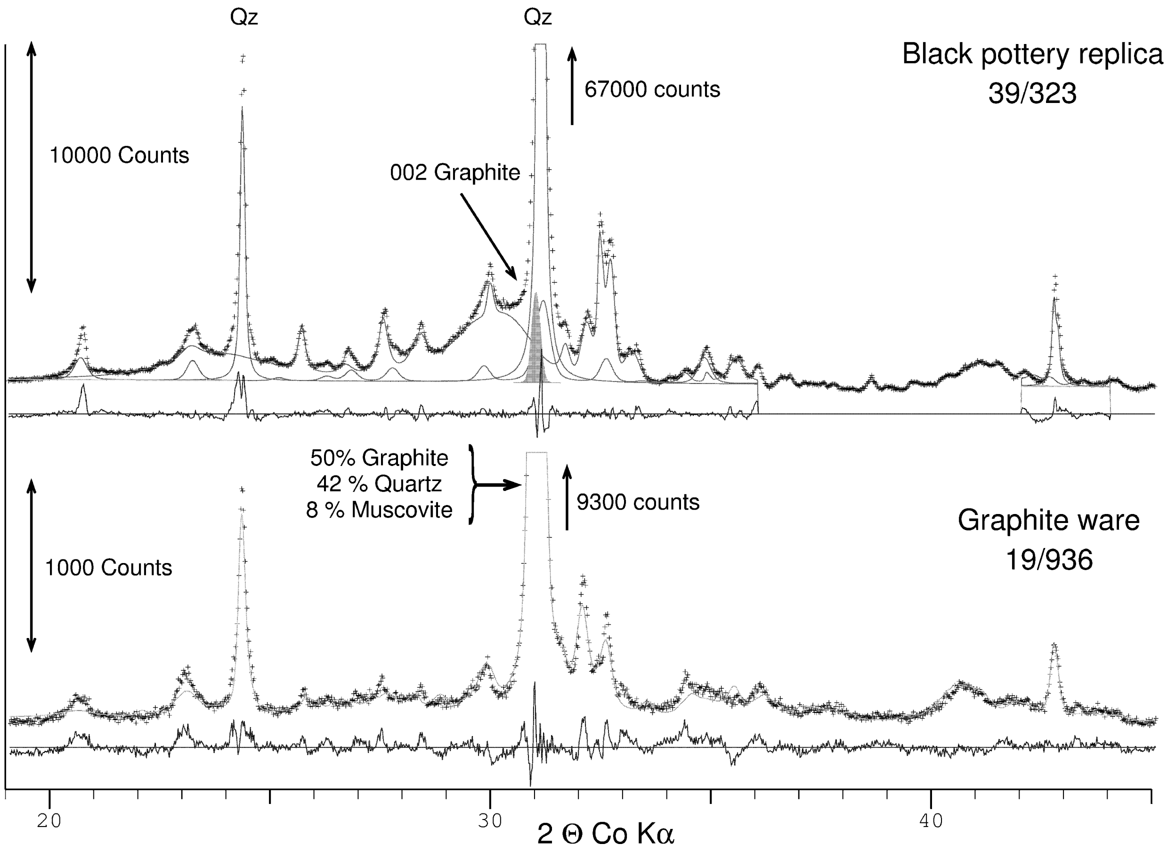

Figure 5. XRD-patterns of a black replica pottery (39/323) and of graphite ware from Manching (19/936). Measured values are shown as +, solid lines represent patterns for the individual single phases (*top*) or the sum of all phases (*bottom*). Residuals are shown below each scan. The graphite 002 peak ($I_{rel} = 100$) in 39/323 is fitted with the MacClayFit program and shown in gray. The plot of the graphite ware (19/936) shows the result of a Rietveld treatment. The contributions of graphite, quartz and muscovite to the peak at 31.1 °2Θ are noted in the plot.

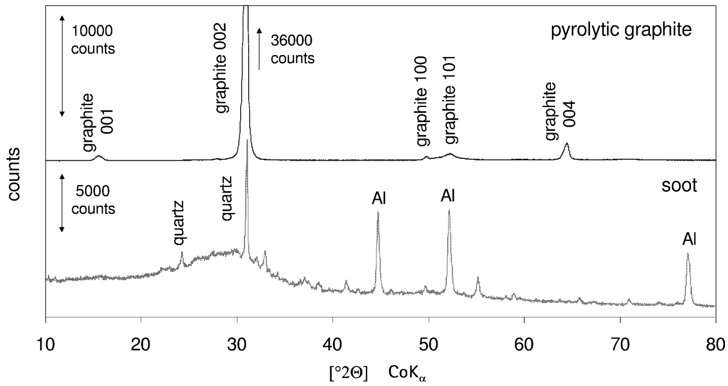

Figure 6. X-ray diffraction patterns of crystalline pyrolytic graphite and of a soot sample from the pipe of a stove in which wood had been burnt. The aluminum peaks (Al) are from the sample holder.

of only 0.04 g/g or, more probably, because most of the carbon in this sample is non-crystalline. Non-crystalline carbon yields only very broad intensity maxima that cannot be detected in the spectra of ceramics. This can be seen in Figure 6 which shows the X-ray diffraction patterns of crystalline pyrolytic graphite and of a soot sample from the pipe of a stove in which wood had been burnt.

3.3. DISCUSSION

The refined graphite contents agree reasonably well with the chemically determined amounts only for the two samples where well crystallised graphite from an ore deposit was added to the clay. Thermal transformation of carbon-containing precursors such as plant materials to carbon yields more or less disordered phases, which exhibit substantial peak broadening. Additionally, carbon is a poor scatterer for X-rays. Both factors constrain the ability of X-ray diffraction to identify and quantify carbon in archaeological samples to specimens which are rich in carbon ($>0.1\ \mathrm{g\,C \cdot g^{-1}}$) and which contain well crystalline graphite. The broad diffraction maximum of non crystalline carbon cannot be detected by common Rietveld programs [13]. Nevertheless, the examples presented here show that, with proper care, quantitative determination of mineral phases like graphite in ceramics are presently within reach. The combination with other, independent methods like chemical carbon analyses and scanning electron microscopy, will increases the reliability with which minor phases in ceramics can be identified and to some extent also quantified.

Acknowledgement

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